

Adrien Trahan

+1 438 523 4409 | cv@adientrahan.com | French & English (C1)
github.com/adientrahan | adientrahan.com

Research Interests

Reinforcement learning, representation learning, and learning-guided search. I am particularly interested in predictive and self-supervised representations (JEPA-style architectures) and in how learned representations and value functions can guide planning and reasoning. I am also open to other directions in machine learning and mathematics.

Education

Polytechnique Montréal

Bachelor's Degree in Software Engineering, Concentration in AI and Data Science

Relevant courses: Machine Learning, Reinforcement Learning, Optimization Methods and Algorithms, Probability and Statistics Methods for AI, Applied Linear Algebra

August 2023 to April 2027

Honours – GPA: 3.84 / 4.0

Politecnico di Torino (Turin, Italy)

2-semester exchange, registered at the Master's level in Computer Engineering

September 2025 to June 2026

29.08 / 30

Research Experience

Task Arithmetic and Federated Learning – Research Project

Politecnico di Torino

2026
github.com/AdrienTrahan/FL-task-arithmetic

- Studied task arithmetic in a federated learning setting and wrote up the findings as a research paper
- Ran large-scale experiments with pretrained vision models such as DINO, totalling 60+ GPU hours

Independent Research

Self-directed

present

- Small-scale experiments with JEPA-style architectures and regularization approaches (sub-JEPA, SIGReg, VISReg) for self-supervised representation learning in MiniGrid environments
- Exploring whether DeepCubeA-style learned heuristic search can be applied to automated theorem proving, framing proof construction as pathfinding in a graph of proof states

Professional Experience

Ivado Labs

Software Engineering Intern (machine learning / data)

May 2025 to August 2025

- Contributed to the outperformance of a trading model on a \$50M+ portfolio on the electricity market by identifying and integrating key new data sources
- Improved the reliability of the model's predictions by uncovering flaws in their distribution and building a data (ETL) pipeline from scratch in Python

Thirdbridge Inc.

Software Development Intern

May 2024 to August 2024

- Delivered the redesign of a \$1M+ contracting company's website, still in production over 2 years later (React, GatsbyJS, Payload CMS)

L3Harris Technologies, Inc.

Business Intelligence Intern (returning intern)

May 2022 to August 2022, May 2023 to August 2023

- Improved work-progress tracking for 10 teams via Power BI dashboards, coordinating the SQL integration of hundreds of tables from 10+ databases

Awards and Distinctions

Computer Science Games (CS Games)

First place in an inter-university competition

April 2024

14 universities, nearly 300 students

Hackatown, Montréal

Audience Choice Award among 50 university teams

January 2020

\$800 in prizes

Polytechnique Hackathon

First place in programming contests

April 2017 and April 2019

\$1,000 in scholarships

Technical Skills

Proficient: PyTorch, Python, Typescript, React, SvelteKit, Swift, MongoDB

Experienced: C++, Java, PostgreSQL, Microsoft SQL Server, Git